

Improving relocalization accuracy via environment prior

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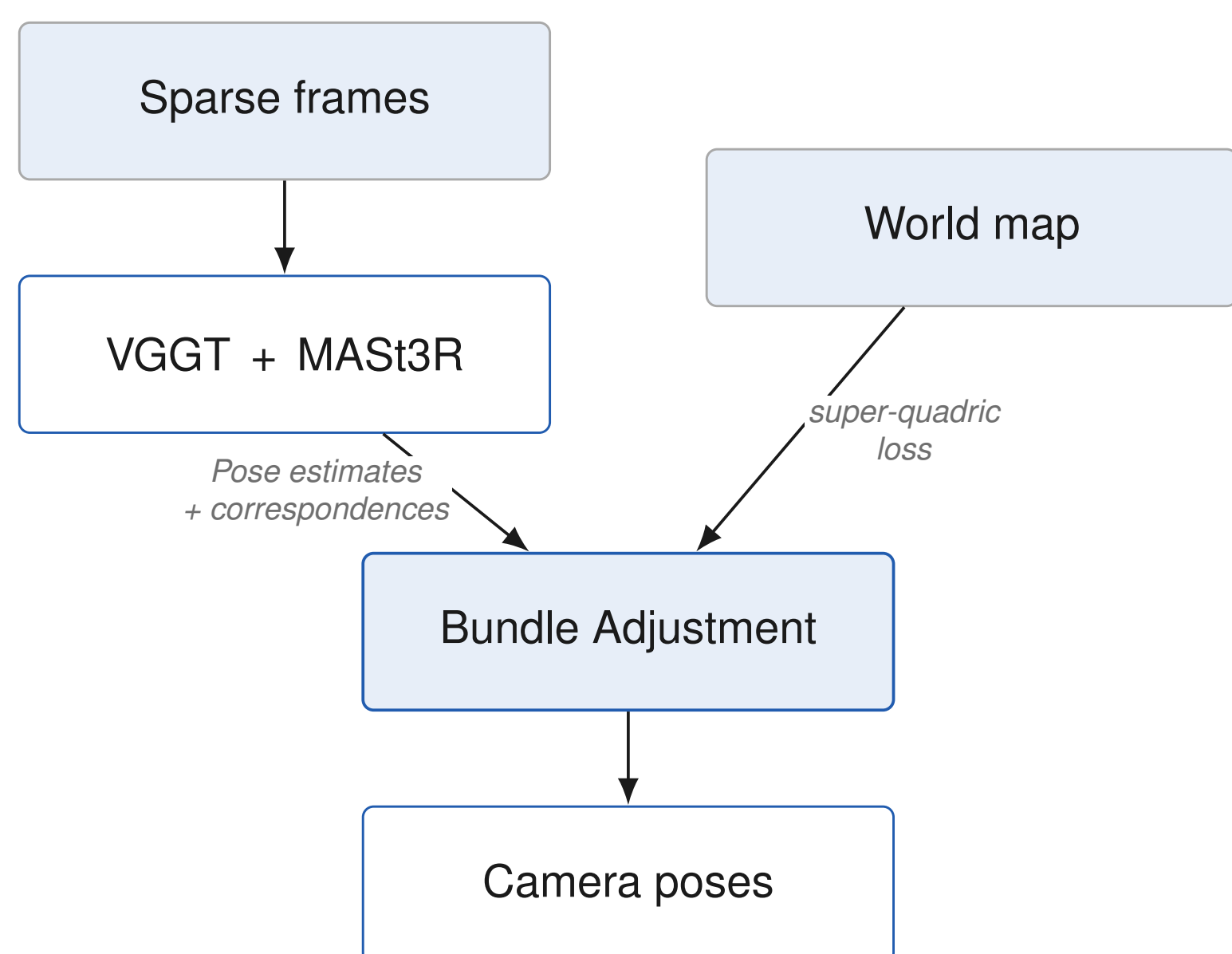
1 Introduction

Recent advancements in transformer-based Structure from Motion (SfM) took the 3D-Vision community by storm [1]. Combining fast inference times with strong accuracy, they outperform traditional feature-matching + bundle adjustment (BA) pipelines in many settings [3].

In relocalization problems such as the “kidnapped robot” scenario, accurate camera pose estimation becomes the main objective while a prior on the environment, such as a point cloud, exists. Current transformer-based SfM approaches cannot exploit such priors.

This work combines transformer-based SfM with such a prior, encoded as a **super-quadric cloud** [4]. Super-quadrics are versatile geometric primitives that give a denser, more compact description of the environment’s topology than raw point clouds.

2 Method Overview



3 The Environment Prior

SuperDec [4] fits a compact set of **super-quadrics** to the scene point cloud. During BA, each triangulated point is associated to its nearest primitive and pulled onto that surface by a one-sided point-to-quadric term, re-associated EM-style every few iterations.

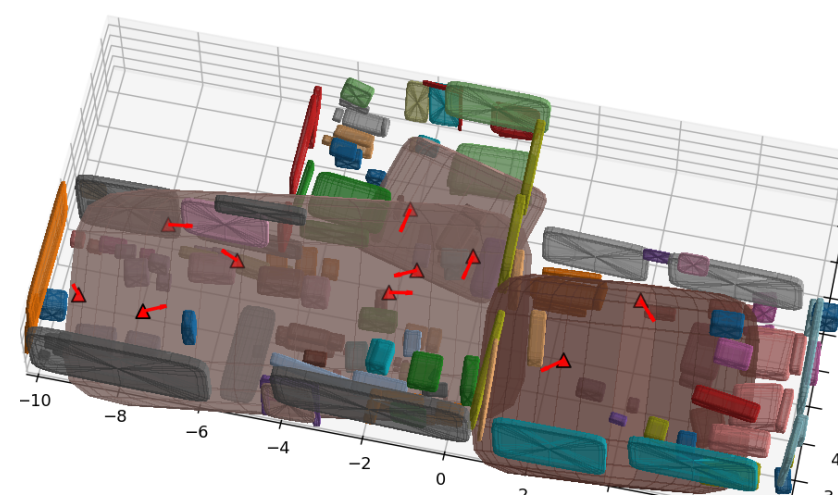


Figure 1: Super-quadric decomposition of a scene (189 primitives) with camera poses.

4 Setup

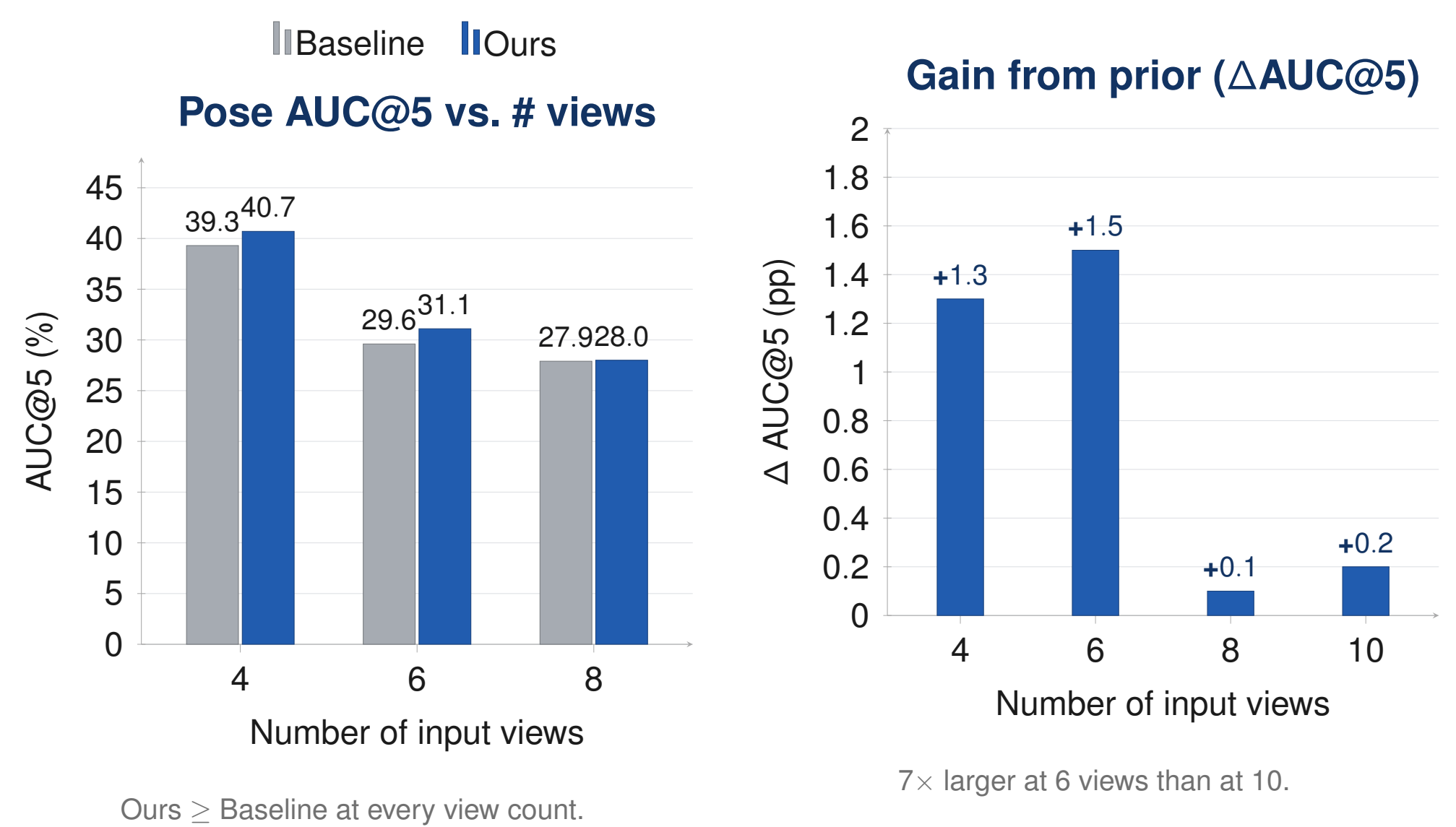
- **Dataset:** Aria Synthetic Environments (10 scenes)
- **Models:** VGGT for feed-forward poses (via MapAnything [2]), MAST3R for matching [3]; SuperDec [4] for the prior
- **Optimization:** full triangulation + surface BA in Ceres

$$E = \underbrace{\sum_{i,j} \rho(\|\pi(R_i, t_i, X_j) - x_{ij}\|^2)}_{\text{reprojection}} + \underbrace{\sum_j \left(\lambda \|q_j\| \cdot \max(0, 1 - F(q_j)^{-\varepsilon_1}) \right)^2}_{\text{one-sided surface prior}}$$

where $F(\cdot)$ is the inside–outside function of the super-quadric $a(j)$ nearest to point X_j , ε_1 its shape parameter, and λ the prior weight [5]. **Metric:** pose AUC@5°.

5 Results

Three configurations: **VGGT-only** (feed-forward poses, AUC@5 = 9.3), **Baseline** (reprojection BA), and **Ours** (BA + the super-quadric prior). **The prior’s benefit grows as the input gets sparser** — the realistic relocalization regime.



+1.5 AUC@5 at 6 views ($\approx$ 5% relative) vs. +0.2 at 10 views

6 Discussion

At high view counts, dense multi-view correspondences already constrain the poses, so the prior is largely redundant (+0.2 at 10 views). As views drop and co-visibility collapses, reprojection alone under-constrains the geometry — and the super-quadric surface fills exactly that gap, recovering up to +2.0 AUC@5 with light tuning at 4 views. ATE stays comparable throughout.

7 Conclusion

Transformer-based SfM cannot natively exploit an environment prior; we close this gap for relocalization. Encoding the prior as a super-quadric cloud and adding a point-to-quadric term to BA improves pose accuracy over both VGGT-only and standard BA — most where it matters: **sparse, low-covisibility views**.

References

- [1] Wang et al., *VGGT: Visual Geometry Grounded Transformer*. CVPR 2025.
- [2] Keetha et al., *MapAnything: Universal Feed-Forward Metric 3D Reconstruction*. 3DV 2026.
- [3] Leroy et al., *Grounding Image Matching in 3D with MAST3R*. ECCV 2024.
- [4] Fedele et al., *SuperDec: 3D Scene Decomposition with Superquadric Primitives*. ICCV 2025.
- [5] Mueller et al., *Reconstructing People, Places, and Cameras*. CVPR 2025.